SMART DIGITAL EXPENSE TRACKER WITH BUDGET PREDICTION

BCA I Semester | Session 2026–2027 | Lab Sheet-09

Submitted By:

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1. Objective

To develop a menu-driven C++ application that allows users to record daily expenses, categorize spending, monitor monthly budgets, receive warnings, and estimate whether the monthly budget may be exceeded.

2. Problem statement

Students often spend money daily but do not maintain a record. This project provides a simple way to save expense details, view spending history, and compare monthly spending with a planned budget.

3. Modules

1. User registration and login
2. Add daily expense
3. View expense history
4. Category-wise expense report
5. Monthly expense report
6. Set monthly budget
7. Budget warning
8. Expense prediction
9. Search expense
10. Delete expense
11. File saving

4. Concepts used

Variables, data types, loops, conditional statements, structures, vectors, functions, strings, searching, sorting, and file handling.

5. Algorithm / logic

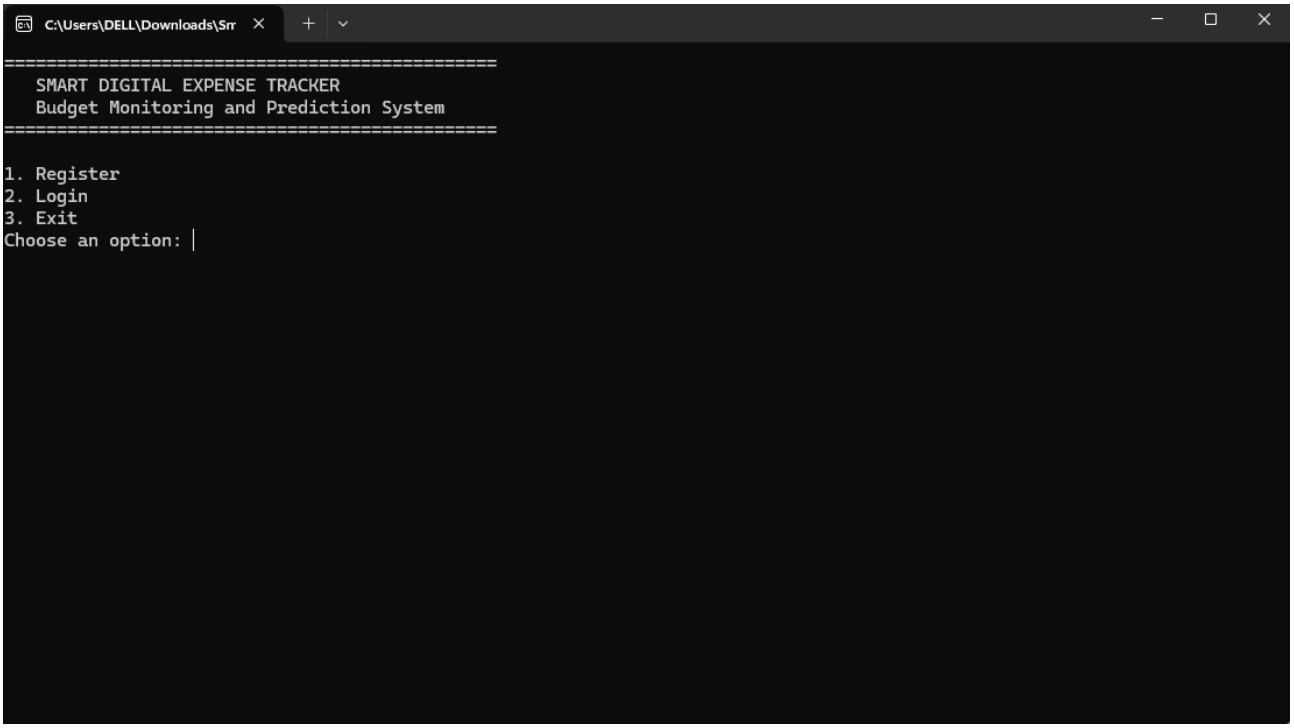
1. Start the program and display the main menu.
2. Register a new user or log in using saved credentials.
3. After login, display the expense menu.
4. Accept expense date, category, description, and amount.
5. Store the expense in a text file with a unique ID.
6. Read saved records to display history and calculate monthly/category totals.
7. Save a monthly budget and compare it with the recorded total.
8. Warn the user when 80% of the budget is used or the budget is exceeded.
9. Estimate monthly spending using average spending per recorded day.
10. Allow searching and deleting records, then exit or log out.

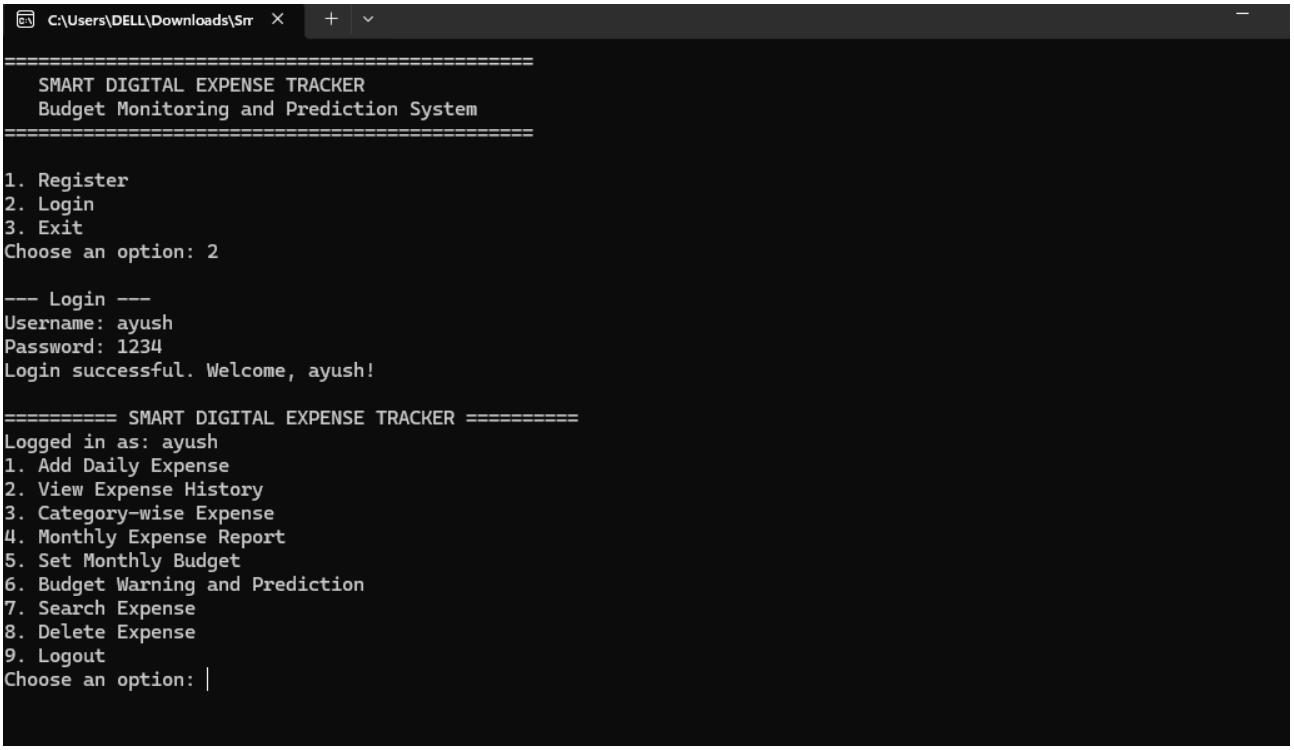
6. Prediction formula

Predicted monthly spending = (Total spending so far ÷ Latest day number recorded) × Total days in the

month.

7. Sample input and output





8. Conclusion

The project demonstrates how C++ structures, functions, loops, searching, sorting, and file handling can be combined to solve a practical expense-tracking problem. It helps users understand their spending and make a simple estimate of month-end expenses.

9. How to compile

```
g++ -std=c++17 -Wall -Wextra main.cpp -o expense_tracker
./expense_tracker    (Windows: .\expense_tracker.exe)
```

Note: This is an educational project. Passwords are stored in plain text and should not be real or reused passwords. The prediction is a simple average-based estimate, not AI.